Standard Photon–Baryon Transport as the Transfer Interface for OPH Sources

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November 13, 2026

Paper release: r2066 **Released:** November 13, 2026

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Abstract

We give the conditional photon–baryon transfer equations for an observer-patch cosmology. A photon record process converges to Liouville transport when its generator approaches null-geodesic flow and its scaled quadratic variation vanishes. Supplying the electromagnetic vertex, polarized Thomson matrix, electron mass, recombination history and cosmological background then gives the standard linear multipole hierarchy, acoustic oscillator, diffusion scale and line-of-sight solution in one brightness convention. Repair exchange and collar stress have separate insertion points. The passive native density process has decaying correlations and does not satisfy the photon-flow identification. The result specifies how a physical source would enter standard transfer; it supplies no primordial amplitude, photon-sector continuum proof or observational likelihood.

1 Branch inputs

We use conformal Newtonian-gauge conventions for the scalar perturbations [1, 2].

Assumption 1.1 (Einstein branch)**.** *A common physical clock and Lorentzian metric satisfy* $G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle$ *with local conserved stress. Identifying* $G_{\mathrm{geom}} = \ell_\star^2$ *and* $\Lambda \ell_\star^2 = 3\pi / N_\star$ *additionally assumes the entropy/area and capacity dictionaries. These require a physical region map identifying the counted quantity with the specified geometric area. Reading* $N_\star$ *from measured* Λ *is a calibration, not a prediction. The smooth Einstein law needs compatible causal, volume and clock limits and tensor-curvature convergence or the small-ball/null-balance identification of the spacetime construction [7].*

Assumption 1.2 (Electromagnetic action and photon pole)**.** *The branch has the Maxwell action and a physical electromagnetic vertex of strength* α*, positive-energy quantization, two transverse states, and a positive-residue massless photon pole in the unbroken Lorentz-vacuum phase. These pole and state properties do not follow from the abstract gauge group. A declared source-map root can be used for* α *only after its current response and renormalization scheme are identified with this vertex. The absolute electron mass* m_e *is an imported physical scale. Finite mass coefficients or conditional charged-lepton ratios alone do not establish it.*

Assumption 1.3 (Cosmological background)**.** *The homogeneous limit of the branch is spatially flat FRW,* $\mathrm{d}s^2 = a^2(\tau)\,[-\mathrm{d}\tau^2 + \delta_{ij}\mathrm{d}x^i\mathrm{d}x^j]$*, with Friedmann dynamics from Assumption 1.1. Scalar perturbations are written in the conformal Newtonian gauge,*

$$\mathrm{d}s^2 = a^2(\tau)\left[-(1+2\psi)\,\mathrm{d}\tau^2 + (1-2\phi)\,\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\right],$$

with conventions of Ma–Bertschinger (1995). Overdots are ∂_τ*. The background densities* $\bar{\rho}_\gamma$ *and* $\bar{\rho}_b$ *are supplied. Finite native class populations, calibrated Gibbs counts and conditional free-gas fractions do not derive these physical cosmological species densities [13].*

Assumption 1.4 (Recombination import)**.** *The free-electron history* $n_e(\tau)$ *(equivalently* $x_e(\tau)$*) is computed from standard atomic physics (Saha/Peebles-class) using imported atomic data. It is in the same import class as the atomic clock unit of the gravity construction: it contains no CMB-target information, so it is a permissible declared input under the anti-fitting rule, but it is an import, not a branch output.*

Assumption 1.5 (Conditional initial spectrum)**.** *Initial curvature perturbations* $\zeta(\mathbf{k})$ *are Gaussian with spectrum* $\Delta^2_\zeta(k) = A_s\,(k/k_*)^{n_s-1}$*. The Gaussian and statistically isotropic character is the maximum-entropy screen-covariance theorem. A finite tilt certificate requires a source generator, the weighted orientation-half identity, and a physical clock receipt. The radial theorem proves that one-shell inversion is nonunique and supplies physical source-dilation and cross-covariance tomography as conditional uniqueness routes. Finite source instantiation is not supplied here. This note consumes* A_s *and* n_s *as gated inputs and derives no initial spectrum. The registered branch* $n_s = 1 - P/48$ *fixes* P *at its inherited* α*-calibrated value, and the executable observer construction does not derive that branch from its joint records [12].*

2 Conditional Refinement Liouville Transport

Definition 2.1 (One-photon record density)**.** *Conditional on the physical pole in Assumption 1.2, let* $\mathcal{E}_\gamma$ *be its positive-energy photon sector. Between record commits, a single excitation is a record continued across patch overlaps. Coarse-graining patch position and overlap momentum labels defines the phase-space record density* $f(x^i, q, \hat{n}, \tau)$ *on the forward mass shell, where* q *is comoving momentum magnitude and* $\hat{n}$ *the direction.*

Theorem 2.2 (Refinement Liouville)**.** *On a branch with the Maxwell action, quantum photon-pole receipt, and overlap Lorentz kinematics from Assumptions 1.2 and 1.1, let* P_r *be the quotient-lumpable free-flow transition kernel at step* h_r*. For every frozen smooth compactly supported test function* ϕ*, set*

$$L_r\phi = \frac{P_r\phi - \phi}{h_r}, \qquad \Gamma_r(\phi) = \frac{P_r(\phi^2) - (P_r\phi)^2}{h_r}.$$

The photon refinement convergence condition consists of compact containment/tightness, convergence of initial laws, uniform-on-compact generator convergence $L_r\phi \to X_H\phi$ *to the null-geodesic Hamiltonian vector field, vanishing quadratic variation* $\Gamma_r(\phi) \to 0$*, uniqueness of the Hamiltonian ODE, refinement and schedule naturality, and vanishing boundary and phase-space-measure defects. Under this receipt, the between-commit record flow satisfies, in the refinement limit,*

$$\frac{Df}{\mathrm{d}\tau} \equiv \frac{\partial f}{\partial \tau} + \frac{\mathrm{d}x^i}{\mathrm{d}\tau}\frac{\partial f}{\partial x^i} + \frac{\mathrm{d}q}{\mathrm{d}\tau}\frac{\partial f}{\partial q} + \frac{\mathrm{d}\hat{n}}{\mathrm{d}\tau}\cdot\frac{\partial f}{\partial \hat{n}} = C[f],$$

where the characteristics are the null geodesics of g_{ab} *and* $C[f]$ *is supported on record commits only. In particular the collisionless flow preserves the Liouville measure.*

Proof sketch under the convergence condition. For every test function, the discrete process minus the time integral of $L_r\phi$ is a martingale whose quadratic variation is controlled by $\Gamma_r(\phi)$. Tightness gives subsequential limits and vanishing quadratic variation makes every limiting martingale deterministic. Generator convergence says that each limit solves the martingale problem for X_H, equivalently the null-geodesic Hamiltonian ODE. Uniqueness identifies the whole limit, independently of the accepted repair schedule. Thus $\mathrm{d}x/\mathrm{d}\tau$, $\mathrm{d}q/\mathrm{d}\tau$, and $\mathrm{d}\hat{n}/\mathrm{d}\tau$ are the geodesic-flow coefficients. On the background of Assumption 1.3, $\mathrm{d}q/\mathrm{d}\tau = q[\dot{\phi} - \ldots]$ has the form used in §4.

Measure preservation. The identified limit is Hamiltonian flow on the mass shell, so it preserves the canonical Liouville measure. The finite measure-defect clause checks that the quotient measures converge to that canonical measure rather than to a different coarse volume form. Accepted repair confluence only removes schedule dependence of committed endpoints. It supplies neither an inverse between-commit dynamics nor a phase-space Jacobian. The orientation doubling of a finite label register is bookkeeping and does not produce either property.

Convergence premise. Photon-sector convergence of the finite patch flow to the displayed generator and zero-quadratic-variation limits is not supplied here. The abstract Lean layer proves the corresponding schedule-independence and refinement-naturality statement in its setting. Section 6 gives the corresponding simulator test. Theorem 2.2 is conditional on this convergence premise. □

Remark 2.3 (Scalar control and photon scope). On the supplied golden population, the positive scalar action $\ddot{q} + c^2Lq + m^2q = 0$ converges at finite times to $u_{tt} - c^2\Delta u + m^2u = 0$ when $\epsilon \to 0$ and $h/\epsilon^2 \to 0$, with an explicit detector bound and a controlled kick–drift–kick discretization. For $m = 0$, its high-frequency Wigner measures follow flat null-geodesic flow [4]. Equating the action speed to the read-law ratio a/Δ makes their limiting cones coincide; this is a common-model condition. The result supplies neither photon polarization, record-process lumpability nor transport on the perturbed metric. Finite Whitney Maxwell evolution preserves constraints on its supplied domain but supplies no photon continuum theorem.

Corollary 2.4 (Collision and diffusion alternatives)**.** *If the finite generator decomposes as* $L_r = L_r^{\mathrm{free}} + C_r$ *and* $C_r\phi \to C\phi$*, then the limiting density obeys*

$$\partial_\tau f + \{f, H\} = C^* f.$$

A nonzero limit of Γ_r *excludes the deterministic Liouville conclusion. A diffusion conclusion additionally requires vanishing jumps and a local second-order limiting generator. A jump limit instead contributes its jump operator to the transport equation.*

For example, on two states let $Q\phi(i) = \phi(1 - i) - \phi(i)$ and $P_h = I + hQ$, $0 < h \leq 1$. For $\phi(i) = i$, the scaled conditional variance is $\Gamma_h(\phi) = 1 - h \to 1$, while the limiting generator Q describes rate-one jumps between the two states. Thus nonzero quadratic variation alone does not identify a diffusion.

Remark 2.5. The transport model places the collision functional at record commits and uses the independently certified measure-preserving flow between them. Claims about entropy production therefore refer to the commit terms counted by $C[f]$. The accepted repair relation and the reversible equilibrium or between-commit dynamics are separate structures.

3 Record Commits and the Thomson Collision Functional

Theorem 3.1 (Scalar-intensity commit-collision functional)**.** *Let commits between photon-class and electron-class records occur with the soft-limit matrix element of the recovered* U(1) *vertex from*

Assumption 1.2. Then, to first order in perturbations and at low occupancy, the collision functional in Theorem 2.2 is the Thomson functional: in the electron rest frame,

$$C[f](q,\hat{n}) \;=\; a\, n_e \sigma_T \left[\frac{3}{16\pi}\int \mathrm{d}\Omega'\,(1+(\hat{n}\cdot\hat{n}')^2)\, f(q,\hat{n}') - f(q,\hat{n})\right], \qquad \sigma_T = \frac{8\pi}{3}\frac{\alpha^2}{m_e^2},$$

boosted to the frame where baryons move with velocity v_b. The source-derived reversible proposal kernel is assumed to obey detailed balance on the angular state space. Write/check orientation labels alone do not imply that property, and the accepted strict-descent repair relation is not the equilibrium kernel. For every fixed photon energy q, isotropic angular distributions are stationary under this Thomson operator. This displayed scalar-intensity kernel does not by itself derive the Stokes-space Thomson matrix needed for polarization.

Proof sketch. At low occupancy the commit rate between momentum cells is bilinear in the occupancies with kernel given by the squared vertex; the soft (photon energy $\ll m_e$) limit of the recovered U(1) vertex is classical Thomson scattering with the stated total cross-section and $1+\cos^2$ angular kernel; this step is standard QED evaluated on the branch couplings and is not repeated here. The detailed-balance receipt equates the stationary weights of each reversible proposal pair. The resulting symmetric rest-frame kernel damps angular anisotropy at each fixed q. It does not mix frequencies, so its stationary kernel contains every isotropic spectrum $f(q)$ rather than a unique Bose or Planck occupancy. Momentum conservation per commit, summed over commits, gives equal and opposite momentum transfer densities to the electron–baryon fluid, with coefficient fixed by $\bar{\rho}_\gamma/\bar{\rho}_b$ bookkeeping; this is the Compton drag used in §4. □

Remark 3.2 (Polarization provenance). The polarization hierarchy used below imports the standard polarized Thomson scattering matrix acting on Stokes brightness. The scalar $1+\cos^2\theta$ kernel displayed in Theorem 3.1 is its intensity projection, not a derivation of the Q/U collision terms. A source-level OPH derivation of the polarized commit kernel from the recovered vertex is not supplied here. The polarization sector and the associated 16/15 diffusion factor are textbook imports rather than consequences of the displayed operator.

Remark 3.3 (Blackbody provenance). The Planck background spectrum is an initial-condition and thermal-history import in this note, not a consequence of the displayed Thomson functional. Even recoil-inclusive Comptonization conserves photon number and generically approaches a Bose–Einstein spectrum with a chemical potential. Selecting the zero-chemical-potential Planck spectrum requires the appropriate early thermalization history, including photon-number-changing channels such as double Compton scattering and bremsstrahlung, or an independently justified blackbody initial condition. An OPH derivation of the blackbody spectrum requires those channels and their rates rather than inferring spectral uniqueness from angular detailed balance.

Remark 3.4 (Provenance of the normalization). α_{root} is an exact declared-map root. Its use as the physical vertex strength is conditional on the three bridge premises in Assumption 1.2. m_e is an import (Assumption 1.2): OPH supplies no source-derived charged-lepton absolute mass, so σ_T's normalization is an explicit import. The recombination history $n_e(\tau)$ is imported under Assumption 1.4.

Remark 3.5 (OPH-specific channels). The declared two-channel continuation permits the following modifications at linear order: (i) a number-changing repair-exchange channel $\Gamma_{\mathrm{rec}}(k,a)$ adds a monopole source and a spectral-distortion term; it enters $C[f]$ additively and requires an active fiber, a physical clock, and a common-parent response; (ii) a collar-packet stress $\rho_A, B_A(k,a)$ enters only the gravitational source side (the Einstein constraints in §4), never $C[f]$. Setting both declared corrections to zero recovers standard transport.

4 The Linear Multipole Hierarchy

Theorem 4.1 (Reduction to the standard hierarchy)**.** *Linearize Theorems 2.2–3.1 on the background of Assumption 1.3 with scalar perturbations, expand the fractional intensity brightness $F_\gamma(\mathbf{k},\hat{n},\tau)$ and polarization brightness G_γ as $F_\gamma = \sum_\ell (-i)^\ell (2\ell+1) F_{\gamma\ell} P_\ell(\mu)$, and likewise for G_γ, with $\mu = \hat{k}\cdot\hat{n}$. Writing $\delta_\gamma = F_{\gamma 0}$, $\theta_\gamma = \frac{3}{4} k F_{\gamma 1}$, $\sigma_\gamma = \frac{1}{2} F_{\gamma 2}$, and $\dot{\kappa} \equiv a n_e \sigma_T$, the system is (Ma–Bertschinger 1995, eq. 64):*

$$
\begin{aligned}
\dot{\delta}_\gamma &= -\tfrac{4}{3}\theta_\gamma + 4\dot{\phi},\\
\dot{\theta}_\gamma &= k^2\Big(\tfrac{1}{4}\delta_\gamma - \sigma_\gamma\Big) + k^2\psi + \dot{\kappa}\,(\theta_b - \theta_\gamma),\\
2\dot{\sigma}_\gamma &= \tfrac{8}{15}\theta_\gamma - \tfrac{3}{5} k F_{\gamma 3} - \tfrac{9}{5}\dot{\kappa}\,\sigma_\gamma + \tfrac{1}{10}\dot{\kappa}\,(G_{\gamma 0} + G_{\gamma 2}),\\
\dot{F}_{\gamma\ell} &= \frac{k}{2\ell+1}\Big[\ell F_{\gamma(\ell-1)} - (\ell+1) F_{\gamma(\ell+1)}\Big] - \dot{\kappa} F_{\gamma\ell}, \qquad \ell \geq 3,\\
\dot{G}_{\gamma\ell} &= \frac{k}{2\ell+1}\Big[\ell G_{\gamma(\ell-1)} - (\ell+1) G_{\gamma(\ell+1)}\Big] + \dot{\kappa}\Big[-G_{\gamma\ell} + \tfrac{1}{2}\,(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2})\Big(\delta_{\ell 0} + \tfrac{\delta_{\ell 2}}{5}\Big)\Big],
\end{aligned}
$$

coupled to baryons,

$$
\dot{\delta}_b = -\theta_b + 3\dot{\phi}, \qquad \dot{\theta}_b = -\frac{\dot{a}}{a}\theta_b + c_b^2 k^2 \delta_b + k^2\psi + \frac{4\bar{\rho}_\gamma}{3\bar{\rho}_b}\,\dot{\kappa}\,(\theta_\gamma - \theta_b),
$$

and to the Einstein constraints of Assumption 1.1,

$$
k^2\phi + 3\frac{\dot{a}}{a}\Big(\dot{\phi} + \frac{\dot{a}}{a}\psi\Big) = -4\pi G a^2\,\delta\rho_{\mathrm{tot}}, \qquad k^2(\phi - \psi) = 12\pi G a^2 (\bar{\rho} + \bar{p})\sigma_{\mathrm{tot}},
$$

where $\delta\rho_{\mathrm{tot}}$ and σ_{tot} include the gated ρ_A, B_A contributions of Remark 3.5. On the standard branch the additional contributions are set to zero. With $\Gamma_{\mathrm{rec}} = B_A = 0$ this is the standard Einstein–Boltzmann system.

Proof. For scalar intensity, insert the geodesic characteristics of Theorem 2.2 on the perturbed metric, the collision functional of Theorem 3.1, keep first order, and project on Legendre multipoles. The $G_{\gamma\ell}$ equations additionally use the imported polarized Thomson matrix stated above. The subsequent calculation is the textbook computation (Ma–Bertschinger 1995; Dodelson, *Modern Cosmology*). The two conditional OPH channels appear at their unique insertion points. □

5 Acoustic normal form, damping, and the line of sight

Proposition 5.1 (Acoustic oscillations)**.** *In the tight-coupling regime $\dot{\kappa} \gg k, \frac{\dot{a}}{a}$, neglect baryon thermal pressure and define $R \equiv 3\bar{\rho}_b/4\bar{\rho}_\gamma$ and $\Theta_0 = \delta_\gamma/4$. To leading order in $1/\dot{\kappa}$,*

$$
\frac{\mathrm{d}}{\mathrm{d}\tau}\Big[(1+R)\,\dot{\Theta}_0\Big] + \frac{k^2}{3}\,\Theta_0 = -\frac{k^2}{3}(1+R)\,\psi + \frac{\mathrm{d}}{\mathrm{d}\tau}\Big[(1+R)\,\dot{\phi}\Big],
$$

a driven oscillator with sound speed $c_s^2 = 1/[3(1+R)]$. Its homogeneous WKB phase, when the coefficients vary slowly compared with an oscillation, is $k r_s$, where

$$
r_s(\tau_*) = \int_0^{\tau_*} c_s\,\mathrm{d}\tau,
$$

at last scattering τ_*. A cosine initial phase gives the approximate unforced extrema $k_n r_s \simeq n\pi$. Baryon loading changes the equilibrium offset and relative compression and rarefaction amplitudes;

evolving gravitational potentials drive the oscillator. Initial phases, free-streaming species, finite visibility and angular projection shift the observed peak positions and heights [5]. The oscillator therefore identifies the acoustic mechanism without fixing exact peaks of a measured TT spectrum.

Proposition 5.2 (Silk damping)**.** *At next order in* $1/\dot\kappa$ *the oscillator acquires the diffusion envelope* e^{-k^2/k_D^2} *with*

$$k_D^{-2} = \int_0^\tau \frac{\mathrm{d}\tau'}{6(1+R)\,\dot\kappa}\left[\frac{R^2}{1+R}+\frac{16}{15}\right],$$

the 16/15 *arising from the standard polarization terms [6] of Theorem 4.1; photon diffusion between commits is the physical mechanism, and its OPH reading is transparent: on scales crossed by a random walk of commit-free transport legs, the record field cannot maintain contrast.*

Proposition 5.3 (Line-of-sight solution)**.** *For* $k>0$*, with visibility* $g(\tau)=\dot\kappa\, e^{-\kappa(\tau,\tau_0)}$ *and an optically thick initial endpoint, the temperature multipoles today, for* $\ell\geq 2$*, are*

$$\Theta_\ell(k,\tau_0)=\int_0^{\tau_0}\mathrm{d}\tau\,\Big\{g\Big[\Theta_0+\psi+\tfrac{\Pi}{4}\Big]\,j_\ell(k(\tau_0-\tau))+\Big(\frac{g\,v_b}{k}\Big)^{\cdot}j_\ell+\frac{3}{4k^2}(g\Pi)^{\cdot\cdot}\,j_\ell+e^{-\kappa}(\dot\psi+\dot\phi)\,j_\ell\Big\},$$

($\Theta_\ell=F_{\gamma\ell}/4$*,* $\Pi=(F_{\gamma 2}+G_{\gamma 0}+G_{\gamma 2})/4$*,* $v_b=\theta_b/k$*; every unlabelled* j_ℓ *has argument* $k(\tau_0-\tau)$*), and*

$$C_\ell^{TT}=4\pi\int\frac{\mathrm{d}k}{k}\,\Delta_\zeta^2(k)\left|\frac{\Theta_\ell(k,\tau_0)}{\zeta(k)}\right|^2.$$

Given the conditional (A_s,n_s) *of Assumption 1.5 and the imports of Assumptions 1.2–1.4, this is the standard linear transfer formula. Agreement of an independent implementation with CAMB at identical inputs and sub-percent tolerances is a separate numerical acceptance test.*

The normalization can be checked directly from the stated hierarchy. Dividing the intensity collision term of Ma–Bertschinger by four gives $-\dot\kappa\Pi P_2(\mu)/2$ and the Doppler term $-i\mu\dot\kappa\theta_b/k$. Along the line of sight, $P_2(\mu)=(3\mu^2-1)/2$ and $\partial_\tau e^{ik\mu(\tau-\tau_0)}=ik\mu e^{ik\mu(\tau-\tau_0)}$. Integration by parts therefore gives $g\Pi/4+3(g\Pi)^{\cdot\cdot}/(4k^2)$ and $(g\theta_b)^{\cdot}/k^2$, respectively. The initial boundary terms vanish by optical thickness, and the observer terms affect only the monopole and dipole. These conventions specify the signs and factors in the Seljak–Zaldarriaga line-of-sight construction [3].

6 Source tests and physical scope

A finite source implementation must verify quotient lumpability, tightness, initial-law convergence, first conditional moments approaching X_H, vanishing scaled quadratic variation, and controlled boundary, measure, mass-shell and schedule defects. These conditions must hold for the photon process itself. A nonzero second-moment limit excludes deterministic Liouville transport; jump sizes and the limiting generator distinguish diffusion from jump transport.

At matched inputs an independent numerical implementation should reproduce temperature and polarization transfer to stated tolerances. That test does not derive its imported electron mass, recombination history, primordial amplitude, radial lift or geometric background. A physical prediction also requires the collision kernel and the same current/clock normalization. Neither a source-map root nor the conditional formula $n_s=1-P/48$ establishes those identifications.

Native repair diffusion and the photon interface. For the specified fair conservative port repair, mean load evolves as $e^{-tL/2}$, with t the native attempt count divided by the seam count, and stationary density has covariance $\kappa e^{-|t-s|L/2}\Pi$. Its integrated readback has a fixed-graph Gaussian limit with inverse-Laplacian covariance; the literal finite-time average has the different normalization $\kappa h(\lambda T/2)$. The exact derivation, local attempted-event record correction and finite-state controls are given in the screen-microphysics companion [8] and preserved in the observer-dynamics evidence archive [14]. Along one common subsequence of refining source screens, actual native density snapshots have joint Gaussian limits with covariance $\exp(-|\tau - u|A)$, where τ is the native attempt count divided by $1260 \cdot 16^l$ and A is a positive self-adjoint operator of heat-trace spectral dimension two on the screen [11, 10]. Coherent waves $\cos(t\sqrt{A})$ require a declared wave processor whose parameter differs from passive waiting, and they propagate at most at speed $1/\sqrt{2}$ in the flat comparison distance per coherent wave parameter [9]. This two-dimensional screen operator supplies no $3 + 1$ null-geodesic flow or photon sector, and its density limit is a stationary Gaussian process, not a deterministic Liouville flow. These statements instantiate a diffusion/record mechanism. They do not verify the vanishing conditional second moment required for a deterministic photon Liouville limit, nor do they provide polarization, Thomson normalization or recombination.

A nondecreasing global change of time keeps every positive mean mode $\exp[-\lambda g(t)/2]$ positive and decreasing. It cannot supply an oscillatory wave propagator. For any current-state centered observable of a stationary reversible continuous-time repair chain, its autocorrelation is a nonnegative sum of decaying exponentials. The corresponding discrete-time conclusion requires a nonnegative spectrum, as in a lazy chain; a nonlazy two-state flip is a counterexample. These restrictions apply to the stated generator and readout, not to extended dynamics with momentum, active forcing, nonstationary preparation or different records. The passive source matching and acoustic initial data used by a conditional CMB transport calculation are additional physical inputs.

Finite collision and expansion controls. A declared quantum bath acts on the calibrated source with 180 fermion modes and 56 boson modes at global occupation cap two. Whole-colour fermion collisions have the form $\Phi_g(X) = X/2 + \tau_g \operatorname{Tr}_g(X)/2$; the capped boson instrument retains creation, removal, hold and forbidden-zero records. Complete Kraus identities give the finite Gibbs fixed point. A two-sweep vacuum replacement subchannel and the fermion collision factors give an explicit reference-inclusive contraction bound for the terminal source marginal. The full bath archive retains its information about the input. This constructive finite collision law, including its supplied temperature and services, is implemented in `model/quantum_thermal_relaxation`; it is not a photon Thomson kernel or a derivation of cosmological thermalization.

On the same fermion mass kernel, the prescribed replacement $p \mapsto p/a(t)$ gives 90 disjoint canonical pairs. Their exact signed two-mode unitaries lift to the full fermion space, preserving pair coherence, all twelve species and complete number records. Outward interval products and a midpoint-evolution comparison bound the continuous cubic ramp from $a = 1$ to $a = 4$. On the supplied initial vacuum, the total mean particle number lies in $[0, 0.2282]$ at duration 8 and in $[2.6040, 2.6109]$ at duration 1, in the declared GeV^{-1} parameter. The sudden quench gives approximately 3.28341935. The massless mode has exactly zero production. These bounds sum local pair errors with their actual multiplicities, as reproduced by `model/expanding_quadratic_particles`. They establish suppressed production for the slower declared ramp. The expansion function, calibrated masses and clock remain inputs; Einstein backreaction and observed cosmic abundances are not inferred.

The hierarchy and its acoustic consequences are standard plasma physics under the stated

imports. A physical OPH CMB prediction additionally needs the source-level photon and polarized collision limits, blackbody thermalization, curvature initial data and radial observation map, and one likelihood with the same background and nuisance conventions. The two additional response terms are absent on the standard branch; a failed source identification does not prove their physical value is zero.

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AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.